No. of Functional Features Included in the Solution

Date	30 JUNE 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Collection	Collects and combines the credit card application and credit history datasets for model development.	Data Preprocessing	Done	High
2	Data Preprocessing	Cleans the dataset, handles missing values, performs feature engineering (Age and Years Employed), and encodes categorical features.	Data Preprocessing	Done	High
3	Machine Learning Model Training	Trains Logistic Regression, Decision Tree, Random Forest, and XGBoost models for credit card approval prediction.	ML Model	Done	High
4	Model Evaluation & Selection	Evaluates models using Accuracy, Confusion Matrix, Precision, Recall, and F1-score, and selects the best-performing model.	ML Evaluation	Done	High
5	Web-Based User Interface	Provides a user-friendly form to enter applicant details for prediction.	Frontend (HTML, CSS)	Done	Medium
6	Prediction Engine	Loads the trained model, scaler, and encoders to predict whether the credit card application is approved or rejected.	Flask Backend	Done	High
7	Result Display	Displays the prediction result (Credit Card Approved or Rejected) to the user in the web application.	Flask Backend & Frontend	Done	Medium
8	Model Persistence	Saves the trained XGBoost model, scaler, and label encoders as .pkl files for reuse in the Flask application.	Model Storage	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6 (Data Collection, Data Preprocessing, Feature Engineering, Model Training, Model Evaluation, Credit Card Prediction)
Additional / Nice-to-Have Features	2 (Flask Web Interface, Model Persistence using Pickle Files)
Features Tested & Verified	8 (All implemented features were tested successfully, including preprocessing, model prediction, and Flask web application.)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Applicant Information Form, Prediction Result Display	HTML form for entering applicant details, Credit Card Approved/Rejected result page
2	Backend / Logic	Data Preprocessing, Feature Encoding, Feature Scaling, ML Prediction, Model Evaluation	Age & Years Employed calculation, Label Encoding, StandardScaler, XGBoost prediction using Flask
3	Database / Storage	Dataset Storage, Model & Encoder Storage	application_record.csv, credit_record.csv, .pkl files for trained model, scaler, and encoders
4	API / Integration	Frontend–Backend Integration, Model Integration	Flask routes (/ and /predict), HTML form submission, loading XGBoost model and returning prediction
5	Security / Authentication	Input Validation	Basic validation for numeric and required input fields in the web form (No user authentication implemented)